

PIA 411 Course Outline

PIA 411 – Portfolio Optimisation and Budget Selection

One budget, many projects, and a solver status that must support the claim.

School	04 – Public Investment
Hours	40 (indicative band 36-45)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	PIA 404 for the ENPV that is being maximised, PIA 401 for the weighted scores this course deliberately does not optimise, and PIA 403 for the capital profiles the budget constrains.
Audience	Officials who assemble an investment programme from a pipeline, and the analysts who have to defend the selection
Pass mark	60 %

What this course is

An appraisal answers whether a project is worth doing. A portfolio answers which projects to do, given a budget – and the second question is not the first repeated. The register makes that concrete in a way no textbook example does.

The project with the largest economic net present value in the pipeline – Northern Corridor Road Upgrade, at 218.5 bn – is selected in none of the 11 runs, under any of the 5 objectives, at either budget envelope. Its capital cost of 1,033.2 bn does not leave room for the smaller projects that together produce more. Ranking by ENPV would put it first; optimisation excludes it.

The course also teaches what a solver actually tells you. All 11 runs terminated OPTIMAL with a gap of 0.0000, so the claim of optimality is supported – which is not always the case and must always be checked. Not one of the 55 budget constraints is recorded as binding, and yet 5 project-rows are deferred for want of budget space. Resolving that apparent contradiction is the heart of the course: in an integer programme a constraint can restrict the answer without being tight, and a mixed-integer programme has no duals to read a shadow price from.

What a graduate can do

Formulate a portfolio selection as a binary integer programme with an objective and an annual budget constraint; distinguish filtering from optimisation and say what each excludes; read a solver's termina-

tion status and say when a result may be called optimal; explain why a non-binding constraint can still restrict an integer solution; say why a mixed-integer programme has no duals and what to do instead; compare the selections produced by different objectives and say when the objective matters; detect an objective that is optimising a field which is NOT AVAILABLE; explain why the highest-value project may be excluded; model a dependency between projects; and write a selection paper that defends the portfolio rather than the ranking.

What this course does not claim

This course does not teach mathematical programming. It teaches enough of it to read a formulation, interpret a solver's output and know when the output does not support the sentence being written about it. Branch and bound, cutting planes and decomposition are elsewhere.

It does not optimise a weighted score. PIA 401 established what a weighted score may decide, and the answer is a long list. This course maximises a quantity with units – an economic net present value, a job count – and treats a score as a filter rather than an objective.

It does not set the budget. The envelope comes from the medium-term expenditure framework and, for a country under a debt sustainability framework, from the borrowing space that framework allows. What an optimiser does with an envelope is the question here; where the envelope comes from is public finance.

It does not resolve the poverty objective. `max_poverty_reduction` is one of the 5 configured objectives, `poverty_effect` is populated on 0 of the 121 project-rows, and every run reports a total poverty effect of zero. The course reports that rather than quietly reinterpreting the result.

Learning outcomes

1. Formulate a portfolio selection as a binary integer programme.
2. Distinguish filtering from optimisation.
3. Read a solver's termination status and gap.
4. Say when a result may be called optimal.
5. Explain why a non-binding constraint can still restrict an integer solution.
6. Say why a mixed-integer programme has no duals.
7. Obtain the value of a relaxed budget without a dual.
8. Compare selections across objectives and say when the objective matters.
9. Detect an objective optimising a field that is NOT AVAILABLE.
10. Explain why the highest-value project may be excluded.
11. Model a dependency between projects and diagnose an infeasibility.
12. Write a selection paper that defends the portfolio.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	Which projects, not whether	1	1.75	–	1	Quiz M01
M02	The formulation	1	4.25	Lab 01	2	Quiz M02

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M03	Filtering is not	1	1.75	–	3	Quiz M03
M04	optimising Reading a solver	1	4.25	Lab 02	4	Quiz M04
M05	Binding, and	1	1.75	–	5	Quiz M05
M06	restricting The shadow price you cannot have	1	4.25	Lab 03	6	Quiz M06
M07	When the objective matters	1	1.75	–	7	Quiz M07
M08	Five objectives, one answer	1	4.25	Lab 04	8	Quiz M08
M09	The objective with no data	1	1.75	–	9	Quiz M09
M10	The project that never gets in	1	4.25	Lab 05	10	Quiz M10
M11	Dependencies and infeasibility	1	4.25	Lab 06	11	Quiz M11
M12	Writing the selection paper	1	1.75	–	12	Quiz M12

Laboratories

Lab	Title	Hours	Deliverable
1	Build the formulation, solve it, and account for the difference	2.5	A working formulation
2	Read every field of a solver record, and write the sentence it licenses	2.5	A solver audit
3	Get the value of a relaxed budget without a dual	2.5	A budget staircase

Lab	Title	Hours	Deliverable
4	Compare objectives, and find out whether one of them does anything	2.5	An objective comparison
5	Defend an exclusion with the counterfactual	2.5	A defended exclusion
6	Add a dependency, add a ceiling, and break the model on purpose	2.5	An extended model

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The formulation, coded and checked	25	Marked on the annual constraint and on separating the filter
A2	The solver record	25	Marked on the three licence conditions
A3	Slack, staircase and objectives	25	Marked on the integrality argument and the coefficient check
A4	The selection paper	25	Marked on the counterfactual and on the exclusions by kind

Final project

A complete portfolio selection paper: the formulation written out with its objective, its decision variables and its constraints; the filter stated separately from the optimisation, with what each excluded and why; the solver's termination status and gap, with a statement of whether the result may be called optimal; the selection under at least two objectives and two budget envelopes, with the projects on which they agree and disagree; an explanation of why the highest-ENPV project is or is not in the portfolio; a constraint report distinguishing a binding constraint from a restricting one; a statement of what a shadow price would tell you and why this model cannot supply one; and a closing paragraph defending the portfolio.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.